Section 6C. Regularization

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Regularization Methods

Instead of looking for a subset of relevant variables, regularization methods constrain the least-squares optimization in order to shrink the coefficient estimates towards zero, resulting in a reduced variance

We have two techniques to regularize our optimization:

- ***Ridge regularization*** (also known as *Tikhonov* regularization), which aims to control the variance of our model to avoid overfitting
- ***The Lasso*** (least absolute shrinkage and selection operator), which apart from controlling the variance, is able to find a subset of relevant input variables

Ridge Regularization

In **Ridge regression**, the coefficients are obtained from the following optimization problem:

$$\widehat{\beta}_{\text{Ridge}}(\lambda) = \arg\min_{\beta} \sum_{i=1}^{N} \left(y_i - f_L(\mathbf{x}_i; \beta)\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

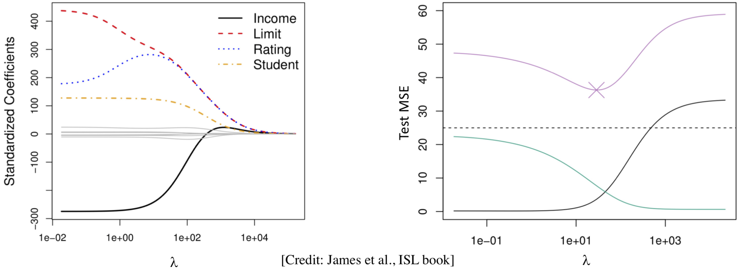
where $\lambda > 0$ is a tuning parameter (to be determined) and $\widehat{\beta}_{\text{Ridge}}$ is a vector of parameters

Comments:

1. Ridge regression seeks to make the RSS $= \sum_{i=1}^{N} \left(y_i - f_L(\mathbf{x}_i; \beta)\right)^2$small
2. At the same time, we have a second term $\lambda \sum_{j=1}^{p} \beta_j^2$, called the *ridge regularization*, aiming to shrink the coefficients β_j towards zero
3. The *tuning parameter* λ controls the relative impact of the ridge regularization term:
   3.1 For $\lambda = 0$, we obtain the standard RSS minimization problem
   3.2 For $\lambda \gg 1$, the regularization term dominates the optimization and $\beta_j \to 0$
4. As λ increases (decreases), the corresponding model is more rigid (flexible). The parameter λ is chosen via cross-validation in order to find the best amount of flexibility

Ridge Regularization: Numerical Example with Credit Dataset

Comment: As shown in the right figure below, as λ increases, the *bias* (black curve) increases but the *variance* (green curve) is reduced. We should choose the value of λ that minimizes the *test MSE* (purple curve), which can be estimated using cross-validation



[Credit: James et al., ISL book]

Issue: No matter how large λ is, ridge regression always includes all the p inputs in the model (i.e., the standardized coefficients β_j are never zero). In other words, ridge regression is not useful to choose a subset of relevant inputs

The Lasso

In contrast with ridge regression, the lasso is an alternative form of regularization that allows us to choose a subset of relevant inputs. The Lasso coefficients are obtained from solving the following optimization problem:

$$\widehat{\beta}_{\text{Lasso}}(\lambda) = \arg\min_{\beta} \sum_{i=1}^{N} \left(y_i - f_L(\mathbf{x}_i; \beta)\right)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

where $\lambda > 0$ is a tuning parameter (to be determined) and $\widehat{\beta}_{\text{Lasso}}$ is a vector of parameters

Comments:

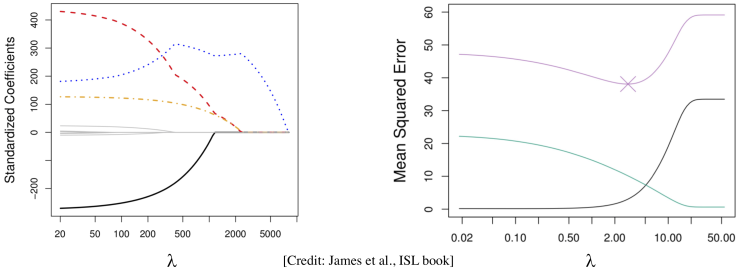
1. In lasso, we have a second term $\lambda \sum_{j=1}^{p} |\beta_j|$, called the *lasso regularization*, aiming to shrink the coefficients β_j towards zero
2. In contrast with ridge regression, lasso is able to force some of the coefficients to be *exactly zero* when λ is sufficiently large. Hence, lasso is able to automatically select a subset of relevant input variables
3. As λ increases (decreases), the corresponding model is more rigid (flexible). The parameter λ is chosen via cross-validation in order to find the best amount of flexibility

The Lasso: Numerical Example

Apply lasso to the **Credit Card** dataset

Comments:

- Left figure: The lasso is able to shrink some of the the standardized coefficients β_j to exactly zero for large enough λ
- Right figure: We should choose the value of λ that minimizes the *test MSE* (purple curve), which can be estimated using cross-validation



[Credit: James et al., ISL book]